

Name:		Index Number:		Class:	
-------	--	---------------	--	--------	--



DUNMAN HIGH SCHOOL

Preliminary Examinations

Year 6

H1 CHEMISTRY

Paper 2 Section A
 Section B

8872/02

24 September 2013
2 hours

Additional Materials: Data Booklet, Graph Paper, Writing Paper

INSTRUCTIONS TO CANDIDATES

- 1 Answer **all** questions in both sections.
- 2 Write your **name** and **class** on this cover page.

Section A

- 3 Write your answers in the spaces provided on this question paper.
- 4 **You are advised to spend a maximum of 60 minutes on Section A.**

Section B

- 5 Write your **name** and **class** on the Cover Sheet provided.
- 6 Write your answers on the separate writing papers provided.
- 7 **Start each question on a fresh sheet of paper.**
**[Marks will be deducted if you fail to do so.]*
- 8 At the end of the examination:
 - Staple or fasten all your work securely together with the Cover Sheet on top.
 - Hand in the question paper separately.
- 9 **You are advised to spend a maximum of 60 minutes on Section B.**

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

You are advised to show all workings in calculations.

You are reminded of the need for good English and clear presentation in your answers.

For Examiner's Use	
Question	Section A
1	11
2	9
3	11
4	9
Total	40

Section A

Answer **all** questions in the spaces provided.

1 (a) (i) Write equations to illustrate the formation of

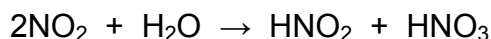
- phosphorus(V) oxide when phosphorus is heated with oxygen

.....

- sulfur(IV) oxide when sulfur is heated with oxygen

.....

(ii) When NO_2 reacts with water, nitrogen undergoes a disproportionation reaction to give a mixture of two acids as shown:



Use this reaction to explain the term *disproportionation*.

.....

.....

.....

(iii) In a similar disproportionation reaction, ClO_2 reacts with aqueous NaOH to produce a solution containing two chlorine-containing sodium salts. Suggest the identities of these two chlorine-containing sodium salts formed.

..... and

[4]

(b) The major source of sulfur for the manufacture of sulfuric acid by the Contact process is the de-sulfurisation of 'sour' natural gas. Many natural gas wells produce a mixture of volatile hydrocarbons (mainly CH_4 and C_2H_6) together with up to 25% hydrogen sulfide, H_2S .

(i) Complete and balance the following equation showing the complete combustion of a gaseous mixture consisting of 2 mol of CH_4 , 1 mol of C_2H_6 and 1 mol of H_2S .

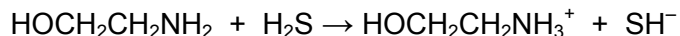


(ii) Suggest why it is important to remove the H_2S before burning the natural gas industrially.

.....

.....

H₂S is removed by passing the 'sour' natural gas through a solvent containing ethanolamine, HOCH₂CH₂NH₂. The following reaction takes place.



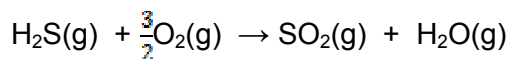
- (iii) State the type of reaction that has occurred.

.....

- (iv) If a sample of natural gas contains 5% by volume of H₂S, calculate the mass of ethanolamine required to remove all the H₂S from a 1000 dm³ sample of gas, measured under room conditions.

The recovered H₂S is converted to sulfur by the following two reactions.

- I Part of the H₂S is burned in air.



- II The gas stream resulting from reaction I is then blended with the remaining H₂S and fed into an iron oxide catalyst bed, where sulfur and water are produced according to the equation.



- (v) Use the following data to calculate ΔH^θ for the reaction between H₂S and SO₂.

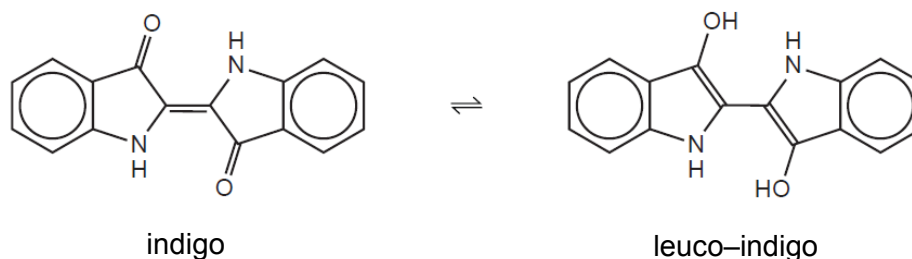
Compound	$\Delta H_f^\theta / \text{kJ mol}^{-1}$
H ₂ S(g)	-21
SO ₂ (g)	-297
H ₂ O(g)	-242
S(g)	+11

[7]

[Total: 11]

- 2 Indigo is the dye used in blue jeans. Although originally extracted from plants of the type *indigofera*, it is now almost entirely made artificially.

Indigo is insoluble in water but this disadvantage can be overcome by converting it into the water-soluble colourless Leuco-indigo. If cloth soaked in a solution of Leuco-indigo is left to dry in the air, the Leuco-indigo is converted into the insoluble blue indigo, which is precipitated out onto the fibres of the cloth.



- (a) On the structures above, circle and name four different functional groups present in either indigo or leuco-indigo, which are not benzene rings

[3]

- (b) Calculate the percentage by mass of carbon in Indigo.

[2]

- (c) (i) What type of reaction is the conversion of Indigo into Leuco-indigo?

- (ii) Suggest a laboratory reagent for this reaction.

[2]

- (d) Suggest the reagents and conditions for a reaction that could be used to distinguish between Indigo and Leuco-indigo by identifying Indigo positively. In your test, state the expected observation for each compound.

Reagents and conditions:

Observations:

[2]

[Total:9]

- 3** Sodium benzoate is a preservative and is able to prevent the growth of bacteria and fungi. It is most widely used in acidic foods such as salad dressings, carbonated drinks, jams and fruit juices.

According to the United States Food and Drug Administration, all food preservatives are limited to an amount of 0.1% by weight. However, The International Programme on Chemical Safety found no adverse effects of sodium benzoate in humans at doses of 647–825 mg/kg of body weight per day.

In acidic foods, sodium benzoate is absorbed into the food cells as benzoic acid. This lowers the intracellular pH of food, creating an environment which fungi and bacteria cannot thrive. In typical fruit juices which have a pH of 3.6, the concentration of benzoic acid was found to be $0.001 \text{ mol dm}^{-3}$. In foods with a pH level of 4.2 to 4.5, a mixture of benzoic acid and sodium benzoate is present. In foods above pH 4.5, sodium benzoate becomes ineffective as a preservative.

Sodium benzoate is prepared by adding benzoic acid to a hot concentrated solution of sodium carbonate until effervescence ceases and all reactants are used up. Appropriate separation techniques can be applied to obtain pure solid sodium benzoate, which is stable upon heating.

An alternative to sodium benzoate is potassium benzoate, which works in exactly the same manner.

- (a)** A typical jar of jam weighs 500 g. Calculate its maximum allowable mass of sodium benzoate as a preservative.

[1]

- (b)** Assuming that an average adult weighs 70 kg, calculate the maximum mass of sodium benzoate a person can intake per day.

[1]

(c) (i) Calculate the concentration of H^+ ions in typical fruit juices.

(ii) Hence, deduce the value of K_a of benzoic acid, assuming benzoic acid is the only acid present.

[2]

(d) (i) State the type of solution formed by the mixture of benzoic acid and sodium benzoate in pH 4.2 to 4.5.

(ii) With the aid of a suitable equation, explain what happens when a small amount of hydrochloric acid was added to this mixture.

[3]

(e) (i) Write the balanced equation for the formation of sodium benzoate from sodium carbonate.

(ii) Suggest a simple method to isolate solid sodium benzoate from the reaction in (e)(i).

[2]

(f) Predict, with explanations, if potassium benzoate has a higher or lower boiling point than sodium benzoate.

[2]

[Total: 11]

4 (a) KMnO_4 and $\text{K}_2\text{Cr}_2\text{O}_7$ are oxidising agents, usually used in acidic solutions.

- (i) State the colour changes you would expect to observe, when each of these reagents are heated with butan-1-ol.

KMnO_4 : to

$\text{K}_2\text{Cr}_2\text{O}_7$: to

- (ii) When heated with butan-1-ol, each of these reagents give rise to different products when different conditions are applied. Suggest how the two products can be obtained by using KMnO_4 and $\text{K}_2\text{Cr}_2\text{O}_7$. Include a balanced equation for each set of reagents and conditions.

Reagents and conditions :

Equation :

Reagents and conditions :

Equation :

[6]

- (b) The following table shows the solubilities of butan-1-ol and but-1-ene in water.

Compound	Mass soluble in 1 dm^3
Butan-1-ol	73.0
But-1-ene	2.21

Explain the above data.

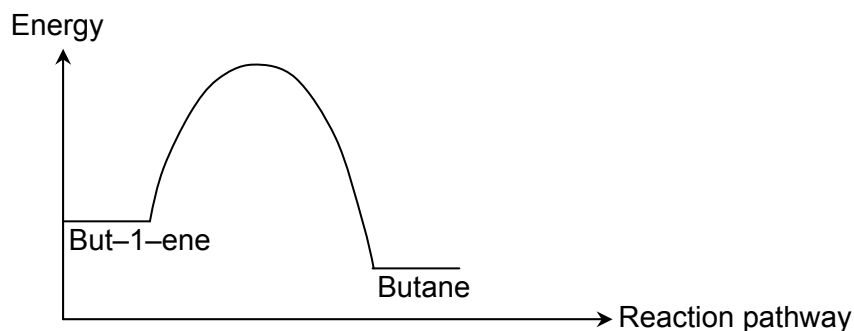
.....

.....

.....

[2]

- (c) The energy profile diagram below represents the hydrogenation of but-1-ene to butane. On the same diagram, draw the energy profile when nickel is added to the reaction.



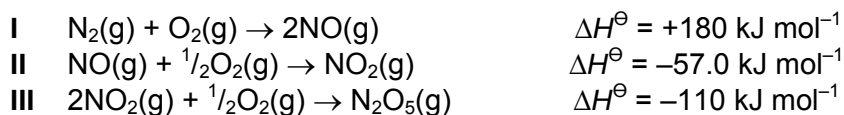
[1]

[Total: 9]

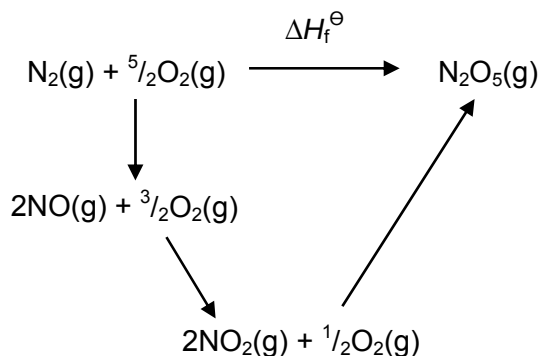
Section B

Answer **two** questions from this section on separate answer paper.

- 5 Dinitrogen pentoxide, N_2O_5 , can be produced by the following reaction sequence in the car engine:



- (a) (i) By using relevant data in the *Data Booklet*, calculate the bond energy of the N=O bond in nitric oxide in reaction I.
- (ii) Suggest an explanation why reaction I occurs in car engine.
- (iii) Nitrogen dioxide is a pollutant that is often produced in car engine. Explain how the pollutant can be removed in the car engine.
- (iv) By using the given energy cycle and the data above, calculate the standard enthalpy change of formation of dinitrogen pentoxide.



[6]

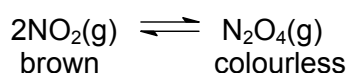
- (b) The rate of reaction for reaction II was investigated and the following data was obtained.

Time/minutes	Experiment 1, with $[\text{NO}] = 0.10 \text{ mol dm}^{-3}$	Experiment 2, with $[\text{NO}] = 0.05 \text{ mol dm}^{-3}$
	$[\text{O}_2] / \text{mol dm}^{-3}$	$[\text{O}_2] / \text{mol dm}^{-3}$
0	0.0050	0.0050
5	0.0031	0.0045
10	0.0019	0.0040
15	0.0011	0.0036
20	0.0007	0.0032
25	0.0005	0.0029
30	0.0004	0.0026

- (i) Explain why nitrogen monoxide is used in large excess.
- (ii) Using the same axes, plot graphs of $[\text{O}_2]$ against time for the two experiments.
- (iii) Use your graphs to determine the order of reaction with respect to O_2 and NO , showing your workings clearly.
- (iv) State the rate equation of reaction II and hence calculate the rate constant, including its units.

[10]

- (c) The nitrogen dioxide produced in reaction II is able to form dinitrogen tetroxide as shown below:



- (i) Draw a dot-and-cross diagram to illustrate the bonding in a nitrogen dioxide molecule and predict its shape.
- (ii) Explain whether the enthalpy change of dimerisation is endothermic or exothermic. Hence, predict the colour change of the reaction mixture when the temperature is increased.

[4]

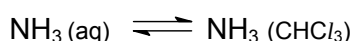
[Total: 20]

- 6 (a)** An experiment is conducted to determine the value of n in the formula of a cation $[\text{Cu}(\text{NH}_3)_n]^{2+}$. This cation is formed when copper(II) reacts with ammonia.

100 cm³ of 0.100 mol dm⁻³ of copper(II) sulfate is mixed with 100 cm³ of aqueous ammonia. The resulting mixture, which contains the cation $[\text{Cu}(\text{NH}_3)_n]^{2+}$ and an excess of ammonia, is then shaken with trichloromethane and allowed to stand for equilibrium to be established. The excess ammonia is dissolved in both the aqueous layer and the trichloromethane layer.

25.0 cm³ of trichloromethane layer is found to neutralise 26.00 cm³ of 0.025 mol dm⁻³ hydrochloric acid while 25.0 cm³ of aqueous layer required 21.00 cm³ of 1.00 mol dm⁻³ hydrochloric acid for neutralisation.

The equilibrium constant, K_c , of ammonia between the aqueous layer and the trichloromethane layer is 25.0.



$$K_c = \frac{[\text{ammonia}]_{\text{aqueous}}}{[\text{ammonia}]_{\text{CHCl}_3}}$$

Calculate:

- (i) the concentration of ammonia in the trichloromethane layer.
- (ii) the concentration of free ammonia in the aqueous layer, using the equilibrium constant.
- (iii) the total concentration of ammonia (free and in the cation, $[\text{Cu}(\text{NH}_3)_n]^{2+}$) in the aqueous layer.
- (iv) the value of n in the formula of the cation $[\text{Cu}(\text{NH}_3)_n]^{2+}$, stating one assumption made in the calculation.

[7]

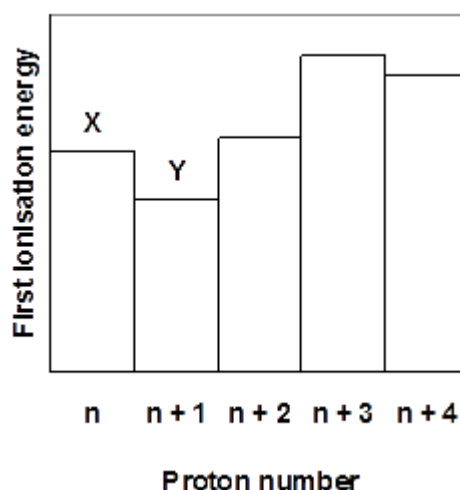
- (b)** Both beryllium difluoride and boron trifluoride can also react with ammonia. However, their mole ratios are different.

Beryllium difluoride and ammonia react in a 1:2 ratio while boron trifluoride and ammonia react in a 1:1 ratio.

- (i) Draw diagrams to show the bonding within these molecules and state the type of bond formed in the products.
- (ii) Explain why the mole ratios are different for these two compounds.

[6]

- (c) The graph below shows the variation in first ionisation energy of some elements in Period 3.



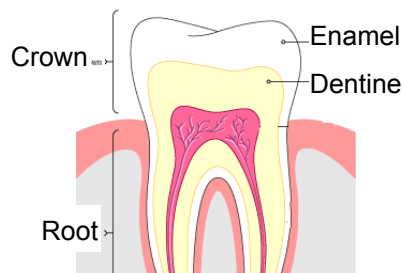
Elements **X** and **Y** have proton numbers n and $n+1$ respectively.

- (i) By considering the trend in first ionisation energy, state and explain which Group element **Y** belongs to.
- (ii) Describe the behaviour of the oxide of **X** in dilute hydrochloric acid and sodium hydroxide respectively.
- (iii) Sodium carbonate was added to a solution of the chloride of **Y** and effervescence was observed. Explain this observation with the aid of relevant equations.

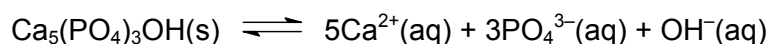
[7]

[Total: 20]

- 7 (a) A tooth is made up of two parts: the crown and the root. Dental crown is the visible part of the teeth which is made of enamel and dentine.



The enamel is made of hydroxyapatite, $\text{Ca}_5(\text{PO}_4)_3\text{OH}$. In the mouth, mineral substances such as calcium ions and phosphate ions that are present in saliva contribute to the formation and decomposition of the hydroxyapatite. These two processes occur simultaneously until an equilibrium is reached. The formation process is called mineralisation of the enamel, whereas the decomposition process is called demineralisation.



Explain how an acidic medium can affect the demineralisation of teeth.

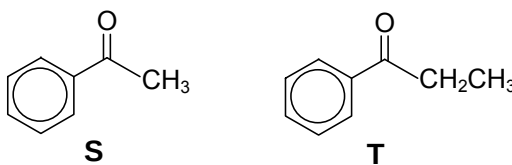
[2]

- (b) An organic compound **R** ($M_r = 113.6$) contains by mass 63.2% carbon, 8.7% hydrogen, and 28.1% oxygen. **R** undergoes an addition reaction with bromine in a 1:1 mole ratio. It also reacts with sodium hydrogencarbonate, liberating carbon dioxide gas.

- Calculate the empirical formula of **R**.
- Determine the molecular formula of **R**.
- Deduce with reasoning, a possible structure for compound **R**.

[5]

- (c) Suggest a suitable chemical test to distinguish between molecules **S** and **T** and describe the expected observations for each molecule.



[2]

- (d) The elements of Group VII, especially chlorine, have played an important role in the development of chemistry. The following describes a reaction involving a chlorine containing compound.

An interhalogen compound **A**, with the formula I_xCl_y , was dissolved in aqueous potassium iodide, and a brown solution was obtained.

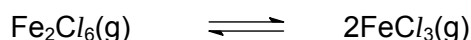


The brown solution was titrated with 0.50 mol dm^{-3} sodium thiosulfate and 34.0 cm^3 was required to reach the end-point. Careful addition of aqueous silver nitrate produced a white precipitate of silver chloride, which weighed 1.83 g when dried.

- (i) Calculate the amount of iodine produced in the reaction shown above.
- (ii) Calculate the amount of chloride ions produced in the reaction shown above.
- (iii) Determine the formula of compound **A** and hence construct a balanced equation between compound **A** and potassium iodide.

[4]

- (e) Anhydrous iron(III) chloride is a dark brown moisture sensitive solid (melting point = 306°C) which forms an acidic solution in water. It exists as a dimer, Fe_2Cl_6 , in the solid state. In the gaseous state, the dimer increasingly dissociates into its monomer until dissociation is complete at 400°C .



0.450 g of Fe_2Cl_6 was placed in a 0.500 dm^3 evacuated flask and heated to 350°C . The amount of gaseous mixture at equilibrium was found to be $2.54 \times 10^{-3} \text{ mol}$.

Using the data above, calculate the following:

- the initial amount of $\text{Fe}_2\text{Cl}_6(\text{g})$;
- the amount of dissociated $\text{Fe}_2\text{Cl}_6(\text{g})$;
- and hence the degree of dissociation of Fe_2Cl_6 at 350°C .

Degree of dissociation = Fraction of dissociated $\text{Fe}_2\text{Cl}_6(\text{g}) \times 100 \%$

[4]

- (f) Iron(III) chloride is a catalyst used in the synthesis of chlorobenzene from benzene.

Explain the catalytic effect of iron(III) chloride on the rate of reaction in terms of the Maxwell-Boltzmann distribution.

[3]

[Total: 20]

BLANK PAGE